

Prove Now, Certify Later: A Calculus for Speculative Theorem Proving

Abstract

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1 Core Proof System

Our core calculus includes the following rules:

$$\frac{\Gamma, P \vdash G}{\Gamma \vdash \bigcirc G} \quad (\bigcirc\text{-intro})$$
$$\frac{\Gamma \vdash \bigcirc G \quad \Gamma \vdash P}{\Gamma \vdash G} \quad (\bigcirc\text{-elim})$$

2 Curry-Howard in CIC

We interpret $\bigcirc$ in CIC based on Partiality Monad (`Part` in Lean) as follows:

$$\bigcirc G = \sum_{P:\text{Prop}} (P \rightarrow G).$$

This interpretation gives soundness for the following typing rules:

$$\frac{\Gamma, x : P \vdash t : G}{\Gamma \vdash \langle P, \lambda x : P. t \rangle : \bigcirc G} \quad (\bigcirc\text{-intro})$$
$$\frac{\Gamma \vdash \langle P, t \rangle : \bigcirc G \quad \Gamma \vdash p : P}{\Gamma \vdash tp : G} \quad (\bigcirc\text{-elim})$$

3 Monad

$\bigcirc$ forms a monad, using the following definitions of `ret` and `bind`:

Proposition 1. $(\bigcirc, \text{ret}, \text{bind})$ is a monad. (i.e., left-unit, right-unit, and associativity holds)

$$\text{ret} : G \rightarrow \bigcirc G$$

$$\text{ret } t := \langle \top, \lambda x : \top. t \rangle : \bigcirc G.$$

$$\text{bind} : \bigcirc P \rightarrow (P \rightarrow \bigcirc Q) \rightarrow \bigcirc Q$$

$$\text{bind } s \, t := \langle \exists p' : \text{fst } s, \text{fst } (t ((\text{snd } s) p')) , \lambda x. \text{snd } (t ((\text{snd } s) (\text{fst } x))) (\text{snd } x) \rangle : \bigcirc Q.$$

In the definition of `bind`, the condition of $\bigcirc Q$ can be understood using $\exists p' : P', Q' \iff P' \wedge Q'$ for non-dependent P' and Q' , which denote conditions for $\bigcirc P$ and $\bigcirc Q$, respectively. Intuitively, `bind` accumulates the conditions.

Proposition 2. $\bigcirc$ satisfies the axioms for lax modality:

$$\text{mult} : \bigcirc \bigcirc P \rightarrow \bigcirc P$$

$$\text{mult } s := \text{bind } s (\lambda p. p) : \bigcirc P.$$

$$\text{mono} : (P \rightarrow Q) \rightarrow (\bigcirc P \rightarrow \bigcirc Q)$$

$$\text{mono } f \, s := \text{bind } s (\lambda p. \text{ret } (f \, p)) : \bigcirc Q.$$

$$\text{strength} : \bigcirc P \rightarrow \bigcirc Q \rightarrow \bigcirc (P \wedge Q)$$

$$\text{strength } s \, t := \text{bind } s (\lambda p. \text{bind } t (\lambda q. \text{ret } \langle p, q \rangle)) : \bigcirc (P \wedge Q).$$

We additionally introduce the following convenient constructor:

$$\text{assume } P : \bigcirc P \quad \text{for any } P : \text{Prop}$$

$$\text{assume } P := \langle P, \lambda x : P. x \rangle : \bigcirc P.$$